PROBABILITY AND STATISTICS I

# Counting Techniques

Lecture 01

Multiplication rule · permutations · combinations

Lecture notes with every worked example,
a model-choosing table, and practice problems for independent work.

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# 1 Why counting comes first

Probability, in the only case where it is easy, is a ratio of two counts. If every outcome of an experiment is equally likely, then

$$P(A) = \frac{\#\text{favourable outcomes}}{\#\text{total outcomes}}.$$

Both numbers in that fraction are counts, which is why a course on probability opens with counting. Nothing in this chapter is hard arithmetic — the whole difficulty is deciding **which count you are being asked for**. Two questions settle it every time:

| Question to ask | If yes | If no |
|---|---|---|
| Does order matter? (is $AB$ different from $BA$?) | permutation — $n!$, $\mathrm{P}_r^n$, or the multi-set form | combination — $\mathrm{C}_r^n$ |
| May an item repeat? (is the item put back?) | multiply the options per step: $n^r$ | the options shrink: $n(n-1)(n-2)\ldots$ |

> **Note.** Answer those two questions **out loud, in words**, before writing a single factorial. Every wrong answer in this chapter is a right formula applied to the wrong model.

Throughout, $n$ is the size of the pool you select from and $r$ is the number of items you select.

# 2 The multiplication rule

**Rule 1.1 — Multiplication rule.**

For a task carried out in $k$ successive steps, if

- there are $n_1$ options (ways) in the first step,
- there are $n_2$ options (ways) in the second step,
- ...
- there are $n_k$ options (ways) in the $k$-th step,

then the total number of ways of completing the task is

$$n = n_1 \times n_2 \times \cdots \times n_k.$$

The rule holds when the steps are **independent choices**: the number of options at each step must not depend on which option was taken earlier. If it does depend, the rule still works provided the **count** is the same whichever earlier option was chosen — which is exactly what happens when you draw without replacement.

## 2.1 Worked examples

**Example 1.** Suppose you have 7 different shirts, 5 different pairs of pants and 3 pairs of shoes. How many outfits are possible?

*Solution.* Three steps — pick a shirt, pick pants, pick shoes:

$$n = n_1 \times n_2 \times n_3 = 7 \times 5 \times 3 = 105 \text{ possible outfits.}$$

**Example 2.** In how many ways can 3 people be selected from a group of 8 people to be president, vice-president and treasurer of the group?

***Solution.*** The three posts are different, so order matters, and nobody holds two posts:

$$n = n_1 \times n_2 \times n_3 = 8 \times 7 \times 6 = 336.$$

$$\boxed{8} \times \boxed{7} \times \boxed{6}$$

**Example 3.** Mohannad is going to assemble a computer by himself. He has the choice of chips from two brands, a hard drive from four, memory from three, and an accessory bundle from five local stores. How many different ways can Mohannad order the parts?

***Solution.*** Since $n_1 = 2$, $n_2 = 4$, $n_3 = 3$ and $n_4 = 5$,

$$n = n_1 \times n_2 \times n_3 \times n_4 = 2 \times 4 \times 3 \times 5 = 120.$$

**Example 4.** In how many ways can a 3-digit number be formed from $\{1, 2, 3, 4, 5\}$ that is less than 400?

***Solution.*** Read the constraint into the **first** slot: the hundreds digit decides whether the number is below 400, so it must come from $\{1, 2, 3\}$, giving $n_1 = 3$. The remaining two digits are unconstrained, and digits may repeat, so $n_2 = 5$ and $n_3 = 5$.

$$\boxed{3} \times \boxed{5} \times \boxed{5}$$

$$n = 3 \times 5 \times 5 = 75.$$

**Watch out.** In Example 4 the digits may repeat, because nothing in the question forbids 111. Whenever a question says "different digits" or "each letter used once", the later slots shrink: $3 \times 4 \times 3 = 36$ instead of 75. One word in the question changes the answer by a factor of two.

**Exercise set 1**

**1.1** A café offers 4 sizes of coffee, 3 types of milk and 6 syrups. How many drinks, taking exactly one of each?

**1.2** A four-digit PIN is chosen with each digit from 0 to 9, repeats allowed. How many PINs?

**1.3** How many 3-digit numbers can be formed from $\{1, 2, 3, 4, 5\}$ that are **even**?

**1.4** How many 3-digit numbers can be formed from $\{1, 2, 3, 4, 5\}$ with no digit repeated?

## 3 Permutations

**Definition 1.1 — Permutation.** A permutation is an arrangement of all or part of a set of objects.

Let us do a permutation problem. How many different arrangements are there for 3 books on a shelf? Books $A$, $B$ and $C$ can be arranged in these ways:

$$ABC \qquad ACB \qquad BAC \qquad BCA \qquad CAB \qquad CBA$$

Six arrangements, or $3! = 3 \times 2 \times 1 = 6$. In a permutation the order of the books **is** important: each different permutation is a different arrangement. The arrangement $ABC$ is different from the arrangement $CBA$, even though they use the same 3 books.

**Definition 1.2 — $n$ factorial.** For any non-negative integer $n$, the quantity called "$n$ factorial" is defined as

$$n! = n \times (n-1) \times \cdots \times 3 \times 2 \times 1,$$

with the special case $0! = 1$.

Examples:

- $3! = 3 \times 2 \times 1 = 6$
- $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
- $8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$

The number of permutations of $n$ different elements is therefore $n!$.

**Note.** $0! = 1$ is a definition, not a computation — there is exactly one way to arrange nothing. It is what makes $\mathrm{C}_n^n = 1$ and $\mathrm{P}_n^n = n!$ come out right, so it earns its keep.

**Example 5.** How many ways are there of arranging 6 items in a display?
***Solution.***

$n = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720.$

## 3.1 Permutations of subsets

In some situations we are interested in the number of arrangements of only **some** of the elements of a set. The number of permutations of the four letters $a$, $b$, $c$, $d$ is $4! = 24$. Now consider the permutations that are possible by taking two letters at a time from four. These would be

$$ab \quad ac \quad ad \quad ba \quad bc \quad bd \quad ca \quad cb \quad cd \quad da \quad db \quad dc$$

Using Rule 1.1 again, we have two positions to fill, with $n_1 = 4$ choices for the first and then $n_2 = 3$ choices for the second, for a total of $n_1 n_2 = 4 \times 3 = 12$ permutations.

**Rule 1.2 — Permutations of subsets.**

The number of permutations of subsets of $r$ elements selected from a set of $n$ different elements is

$$\mathrm{P}_r^n = \frac{n!}{(n-r)!} = n \times (n-1) \times (n-2) \times \cdots \times (n-r+1).$$

**Note.** The right-hand form — $r$ descending factors starting at $n$ — is the one to use by hand. It is the slot picture of Rule 1.1 with the pool shrinking by one each step, and it never asks you to evaluate a factorial you are about to cancel.

**Example 6.** How many ways are there of selecting 2 different items from a group of 6 if order is taken into account?
***Solution.***

$$\mathrm{P}_r^n = \mathrm{P}_2^6 = \frac{6!}{4!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1} = 30.$$

**Example 7.** How many ways are there of selecting a committee of four from a staff of 20 if the order of selection is significant?
***Solution.***

$$\mathrm{P}_r^n = \mathrm{P}_4^{20} = \frac{20!}{16!} = \frac{20 \times 19 \times 18 \times 17 \times 16!}{16!} = 116280.$$

**Example 8.** How many permutations of 3 different digits are there, chosen from the ten digits 0 to 9 inclusive?
***Solution.***

$$\mathrm{P}_r^n = \mathrm{P}_3^{10} = \frac{10!}{7!} = \frac{10 \times 9 \times 8 \times 7!}{7!} = 720.$$

## 3.2 Permutations of similar objects

When some of the objects are indistinguishable, the plain $n!$ over-counts: swapping two identical letters produces the same word twice. Divide out the arrangements you cannot tell apart.

**Rule 1.3 — Permutations of similar objects.**

The number of permutations of $n = n_1 + n_2 + \cdots + n_k$ objects, of which $n_1$ are of one type, $n_2$ are of a second type, ..., and $n_k$ are of a $k$-th type, is

$$\binom{n}{n_1, n_2, \ldots, n_k} = \frac{n!}{n_{1!} \times n_{2!} \times \cdots \times n_{k!}}.$$

**Example 9.** In how many ways can one arrange 2 maths, 3 physics, 3 chemistry and 1 Arabic book on a shelf, if books of the same subject are indistinguishable?
***Solution.*** The total number of possible arrangements is

$$\binom{9}{2, 3, 3, 1} = \frac{9!}{2! \times 3! \times 3! \times 1!} = 5040.$$

**Example 10.** How many different letter arrangements can be made from the letters in the word "STATISTICS"?
***Solution.*** Count the letters first: S appears 3 times, T appears 3 times, I appears 2 times, and A and C once each — $3 + 3 + 2 + 1 + 1 = 10$ letters in all. So

$$\binom{10}{3, 3, 2, 1, 1} = \frac{10!}{3! \times 3! \times 2! \times 1! \times 1!} = 50400.$$

**Watch out.** Tally the repeated letters on paper before dividing. The commonest lost mark in this rule is a letter counted once that appears twice — and the check is free: the $n_i$ must add up to $n$.

### Exercise set 2

**2.1** Evaluate $\mathrm{P}_3^7$ and $\mathrm{C}_3^7$, and say in one sentence why one is six times the other.

**2.2** How many permutations of 4 different letters are there, chosen from the 26 letters of the alphabet?

**2.3** How many different letter arrangements can be made from the letters in "PROBABILITY"?

**2.4** In how many ways can 8 runners finish a race with first, second and third place awarded?

**2.5** How many arrangements of the word "SUCCESS" begin with S?

# 4 Combinations

When the order of selection does not matter, each set of $r$ items has been counted $r!$ times by $\mathrm{P}_r^n$ — once for every way of arranging it. Dividing by $r!$ gives the count of **subsets**.

**Rule 1.4 — Combinations.**

The number of combinations — subsets of $r$ elements that can be selected without regard to order from a set of $n$ elements — is denoted $\binom{n}{r}$ or $\mathrm{C}_r^n$, and

$$\binom{n}{r} = \mathrm{C}_r^n = \frac{n!}{r!(n-r)!}, \quad 0 \le r \le n.$$

**Note.** Relation worth writing on the board once and never re-deriving:

$$\mathrm{P}_r^n = r! \times \mathrm{C}_r^n.$$

Combinations **choose**; permutations **choose and then arrange**. Also $\mathrm{C}_r^n = \mathrm{C}_{n-r}^n$ — choosing the 5 who play is the same as choosing the 5 who sit out, which is often the cheaper computation.

**Example 11.** To play a particular card game, each player is dealt five cards from a standard deck of 52 cards. How many different hands are possible?

***Solution.*** A hand is a set — the order in which the cards arrive is irrelevant:

$$\mathrm{C}_5^{52} = \frac{52!}{5!\ 47!} = \frac{52 \times 51 \times 50 \times 49 \times 48 \times 47!}{5 \times 4 \times 3 \times 2 \times 1 \times 47!} = 2\ 598\ 960.$$

**Example 12.** A student must answer 3 out of 5 essay questions on a test. In how many different ways can the student select the questions?

***Solution.***

$$\mathrm{C}_3^5 = \frac{5!}{3!\ 2!} = \frac{5 \times 4 \times 3!}{3! \times 2 \times 1} = 10.$$

**Example 13.** A basketball team consists of two centers, five forwards and four guards. In how many ways can the coach select a starting line-up of one center, two forwards and two guards?

***Solution.*** Choose within each position, then multiply — Rule 1.4 inside Rule 1.1:

$$\text{Center:} \quad \mathrm{C}_1^2 = \frac{2!}{1!\ 1!} = 2,$$

$$\text{Forwards:} \quad \mathrm{C}_2^5 = \frac{5!}{2!\ 3!} = 10,$$

$$\text{Guards:} \quad \mathrm{C}_2^4 = \frac{4!}{2!\ 2!} = 6.$$

Thus the number of ways to select the starting line-up is

$$C_1^2 \times C_2^5 \times C_2^4 = 2 \times 10 \times 6 = 120.$$

**Example 14.** How many ways are there to choose a committee of 4 persons from a group of 10 persons, if order is not important?
***Solution.*** Here $n = 10$ and $r = 4$, so the number of ways is

$$C_4^{10} = \frac{10!}{4!\ 6!} = 210.$$

### 4.1 Committees with a distinguished member

Two variants of Example 14 appear constantly, and they are not the same question.

**Example 15.** How many ways are there to choose a committee of 4 from 10 people if one of the four is to be the chairperson?
***Solution.*** Split the task into two steps. Choose the chairperson first — 10 ways — then choose the remaining 3 members from the 9 people left:

$$10 \times C_3^9 = 10 \times 84 = 840.$$

Equivalently, choose the committee and then pick its chair from inside it: $C_4^{10} \times 4 = 210 \times 4 = 840$. The two routes must agree, and checking that they do is a good habit.

**Example 16.** Jones is the chairman of a committee. In how many ways can a committee of 5 be chosen from 10 people given that Jones must be one of them?
***Solution.*** Jones is **already chosen**, so there is nothing to decide about him. Fill the other 4 seats from the remaining 9 people:

$$C_4^9 = 126.$$

**Watch out.** "One of them is to be the chair" (Example 15) means a choice is still to be made — multiply. "Jones must be on the committee" (Example 16) means a seat is already spent — shrink both $n$ and $r$ by one. Students routinely answer 15 with $C_4^{10}$ and 16 with $C_5^{10}$; both are wrong for the same reason, a constraint that was read but not used.

## 5 Choosing the model

| Situation | Count | Typical wording |
|---|---|---|
| Successive independent choices | $n_1 \times n_2 \times \cdots \times n_k$ | "one of each", "then" |
| Arrange all $n$ distinct objects | $n!$ | "in how many orders" |
| Arrange $r$ of $n$ distinct objects | $P_r^n = \frac{n!}{(n-r)!}$ | "order matters", "president, VP, treasurer" |
| Arrange $n$ objects with repeats | $\frac{n!}{n_1!\cdots n_k!}$ | "letters of the word …" |
| Choose $r$ of $n$, order irrelevant | $C_r^n = \frac{n!}{r!(n-r)!}$ | "committee", "hand", "sample" |

| Situation | Count | Typical wording |
|---|---|---|
| $r$ slots, $n$ options, repeats allowed | $n^r$ | "repeats are allowed", "with replacement" |

**Note.** Verify every count on a tiny case by listing it. If a formula claims 12 two-letter words from $\{a, b, c, d\}$, write the 12 out once — as in §3.1 — and you will never mistrust the formula again. This is also the fastest way to catch an $\mathrm{P}_r^n$ that should have been a $\mathrm{C}_r^n$: the listed items will contain both $ab$ and $ba$, or they will not.

### Exercise set 3

**3.1** How many different committees of 5 people can be chosen from 10 people?

**3.2** A password consists of two letters of the alphabet followed by three digits from 0 to 9, with repeats allowed. How many passwords are there?

**3.3** A special password consists of four **different** letters of the alphabet, each used once. How many passwords are there?

**3.4** From 6 men and 4 women, how many committees of 4 contain exactly 2 women?

**3.5** From 6 men and 4 women, how many committees of 4 contain **at least** one woman?

**3.6** A bag holds 5 red and 3 blue balls. Three are drawn together. In how many ways can exactly two be red?